Quanta 2: Indeterminacy

The world is a sequence of granular quantum events. These are discrete, granular and individual; they are individual interactions of one physical system with another. An electron, a quantum of a field or a photon does not follow a trajectory in space but appears in a given place and at a given time when colliding with something else. When and where will it appear? There is no way of knowing with certainty. Quantum mechanics introduces an elementary *indeterminacy* to the heart of the world. The future is genuinely unpredictable. This is the second fundamental lesson learned with quantum mechanics.

Due to this indeterminacy, in the world described by quantum mechanics, things are constantly subject to random change. All the variables 'fluctuate' continually, as if, at the smallest scale, everything is constantly vibrating. We do not see these omnipresent fluctuations only because of their small scale; they cannot be observed at a large scale, as when we observe macroscopic bodies. If we look at a stone, it stays still. But if we could see its atoms, we would observe them constantly spread here and there, and in ceaseless vibration. Quantum mechanics reveals to us that, the more we look at the detail of the world, the less constant it is. The world is not made up of tiny pebbles. It is a world of vibrations, a continuous fluctuation, a microscopic swarming of fleeting micro-events.

The atomism of antiquity had anticipated also this aspect of modern physics: the appearance of laws of probability at a deep level. Democritus assumed (just like Newton) that the movement of atoms was rigorously determined by their collisions. But his successor, Epicurus, corrects the determinism of the master and introduces into atomism the notion of indeterminacy – in the same way in which Heisenberg introduces indeterminacy into Newton's determinism. For Epicurus, atoms can on occasion deviate by chance from their course. Lucretius says this in beautiful words: this deviation occurs '*incerto tempore … incertisque loci*':[3] at an uncertain place, at an uncertain time. The same randomness, the same appearance of probability at an elementary level, is the second key discovery about the world that quantum mechanics expresses.

So, how do we compute the probability that an electron in a certain initial position A will reappear, after a given time, in one or another final position B?

In the 1950s, Richard Feynman, who I've already mentioned, found a suggestive method of making this calculation: consider *all* possible trajectories from A to B, that is to say, all possible trajectories the electron can follow (straight, curved, zigzagging …). Each trajectory determines a number. The probability is obtained from the sum of all these numbers. The details of this calculation are not important: what matters is the fact that all trajectories